Table of Contents

1.0 Introduction 2
2.0 Tool Selection and Discussion 2
2.1 FTK Imager 2
2.2 Autopsy 4
2.3 Volatility3 6
3.0 Evidence Acquisition 8
3.1 HDD Imaging 8
3.2 RAM Capture 13
4.0 Evidence Analysis 14
4.1KMSAuto Net.exe 14
4.2 DLLInjector v2.exe 18
4.3 wap.exe 21
5.0 Technical Report 24
5.1 Analysis of KMSAuto Net.exe 24
Process Details: 24
File Hashes: 24
VirusTotal Analysis: 24
Behaviors: 24
5.2 Analysis of DLLInjector v2.exe 25
Process Details: 25
File Hashes: 25
Behaviors: 25
5.3 Analysis of wap.exe 25
Process Details: 25
File Hashes: 25
Behaviors: 25
Conclusion 26
6.0 Conclusion 26
7.0 References 28

1.0 Introduction

In the ever-changing world of cybersecurity threats, forensic investigation must discover and avert harm. Complex digital ecosystems require qualified forensic investigators to analyze broken systems. This article discusses data collecting, analysis, and reporting to discover system malware and dangerous actions. A Windows 10 PC is thoroughly examined for compromise.

This investigation will analyze the victim's computer's unusual activities to establish its cause. We'll describe how data was taken from the compromised machine, how forensic tools were utilized to analyze the case's intricacies, how RAM and HDD data were retrieved, how photos were analyzed for evidence, and how we ended our investigation with a thorough technical report. This technique seeks to identify and eliminate the threat and suggest ways to improve system security and prevent future events.
This document describes forensic investigative methodologies, instruments, and findings. We gather, analyze, and publish data methodically to provide stakeholders clarity and actionable insights for security breach response. Digital forensics and collaboration secure digital assets and mitigate cyber hazards to companies and people.

2.0 Tool Selection and Discussion

2.1 FTK Imager

FTK imager, developed by AccessData, is a prominent tool in the realm of digital forensics. It makes it possible for forensic investigators to produce forensic photographs of memory cards, USB drives, and hard disks. With no modifications to the original evidence, these photographs are precise replicas of the original storage media, retaining all information and data for analysis.



Functions of FTK Imager

1. Live Imaging:

By supporting live imaging, FTK Imager enables investigators to take forensic pictures of operational systems without having to shut them down. This feature is very helpful in situations where the system cannot be turned off.

2. Acquiring Volatile Memory:

Investigative tools can benefit from the important information found in volatile memory, such as RAM. Forensic experts can retrieve and examine volatile memory contents for possible evidence using FTK Imager.

3. Imaging Network Shares and Remote Devices:

Imaging of faraway devices and network shares is made easier with FTK Imager. By using this tool, investigators may broaden the scope of their investigations by gathering evidence from networked sources.

4. Extracting Individual Files and Folders from Forensic Images:

To extract certain files and directories from forensic photos, investigators might utilize FTK Imager. This concentrated investigation makes it possible to examine pertinent evidence in more detail.

Benefits of FTK Imager

1. User-Friendly Interface:

FTK Imager boasts a user-friendly interface that is intuitive and easy to navigate. This feature makes it accessible to both beginners and experienced forensic professionals.

2. Versatility:

FTK Imager offers versatility in managing diverse kinds of proof by supporting an extensive array of file formats and storage media. Because of its adaptability, it is more beneficial in many research situations.

3. Reliability and Accuracy:

FTK Imager ensures data integrity through its verification and validation mechanisms. Its robust algorithms and processes contribute to accurate and reliable forensic imaging.

4. Compatibility with Other Forensic Tools:

With FTK Imager's smooth integration with other well-known forensic tools, investigators may use a variety of tools and methods to improve their analysis and inquiries.

5. Time Efficiency:

FTK Imager offers rapid imaging and analysis capabilities, saving valuable time during investigations. Its efficient processes contribute to streamlined workflows and increased productivity.

2.2 Autopsy

Autopsy, an open-source digital forensics programme, helps investigators find evidence in digital data. Autopsy's capabilities allow forensic photography of storage media, file analysis, keyword searches, timeline reconstruction of file activity, and digital artefact examination. The platform's modular architecture lets investigators add capabilities as required and write detailed reports. Autopsy, compatible with many operating systems and supported by a large user community, is a flexible and accessible tool for digital forensic investigations.



Functions of Autopsy:

Disk Imaging:

Autopsy allows for the creation of forensic images of storage media, preserving the original evidence for analysis without alteration.

File Analysis:

It enables the examination of file metadata, content, and attributes to identify potential evidence relevant to the investigation.

Keyword Search:

Autopsy includes powerful search functionality, allowing investigators to search for specific keywords or patterns within the analyzed data, aiding in the discovery of relevant evidence.

Timeline Analysis:

It provides a timeline view of file activity, showing when files were created, modified, or accessed, assisting in reconstructing the sequence of events.

Artifact Analysis:

Autopsy supports the analysis of various digital artifacts such as internet history, email, chat logs, and registry entries, helping uncover additional evidence.

Benefits of Autopsy:

Open Source:

Autopsy is an open-source tool, freely available to forensic investigators, fostering collaboration and community-driven development.

Scalability:

It can handle large volumes of data efficiently, making it suitable for investigations involving extensive digital media.

Modularity:

Autopsy's modular design allows for the integration of additional modules and plugins, expanding its functionality to meet specific investigative needs.

Ease of Use:

Despite its powerful features, Autopsy offers a user-friendly interface, making it accessible to both novice and experienced investigators.

Cross-Platform Compatibility:

Autopsy is compatible with multiple operating systems, including Windows, Linux, and macOS, ensuring flexibility in deployment across different environments.

Community Support:

Being widely adopted in the digital forensic community, Autopsy benefits from a large user base and active community support, providing resources, documentation, and assistance to users.

2.3 Volatility3

Digital forensics defines volatility as the temporary characteristic of data stored in volatile memory like RAM, which vanishes when the power is off or the machine is turned off. Transitory data often stores critical information including active processes, network connections, and user activity. Tools like the Volatility Framework can extract and analyse this data to reveal a system's state over time. Interpreting volatile memory helps investigators recreate events, find evidence, and find digital investigation-related facts. Forensic investigators who want to study digital systems deeply must understand and use volatility analysis.



Functions of Volatility:

Memory Analysis: Volatility enables the extraction and analysis of volatile memory data from live systems or memory dumps, revealing crucial information such as running processes, network connections, and loaded modules.

Process Analysis: It allows for the examination of running processes, including their attributes, memory allocations, and associated threads, aiding in the identification of malicious or suspicious activities.

Artifact Extraction: Volatility can extract various artifacts from memory dumps, including registry hives, file handles, and cached data, providing valuable evidence for forensic investigations.

Malware Detection: The framework includes plugins for detecting and analyzing malware-related artifacts in memory, assisting investigators in identifying and analyzing malicious software.

Benefits of Volatility:

Timely Analysis: Volatility facilitates rapid memory analysis, enabling investigators to quickly assess the state of a system and identify potential threats or evidence.

Comprehensive Insights: By analyzing volatile memory, Volatility offers comprehensive insights into system activity, helping investigators reconstruct events and understand the behavior of a compromised system.

Non-Intrusive Examination: Volatility allows for non-intrusive examination of live systems without modifying or interrupting their operation, ensuring minimal disruption to ongoing activities.

Forensic Soundness: The framework follows forensic best practices, ensuring the integrity and reliability of analysis results, which are admissible in legal proceedings.

Community Support: With a large and active user community, Volatility benefits from continuous development, support, and the availability of additional plugins and extensions, enhancing its functionality and versatility.

Tools used

FTK IMAGER : Used for HDD Imaging and RAM capturing

AUTOPSY: Analyzing forensic evidence

VOLATILITY3: Analyzing RAM to extract evidence

3.0 Evidence Acquisition

3.1 HDD Imaging

Making an identical, bit-for-bit copy of a hard drive's contents—a process known as hard disc drive (HDD) imaging is a crucial step in digital forensics. By following this process, investigators may obtain a duplicate copy of the data for examination without compromising the original data's integrity. Hard drive imaging plays a crucial role in several scenarios, such as digital inquiries, data retrieval, and guaranteeing the validity of evidence in court cases.

First Step: Start the FTK imager. Go to the top menu of the FTK Imager interface and choose "File". Decide to "Create Disc Image."

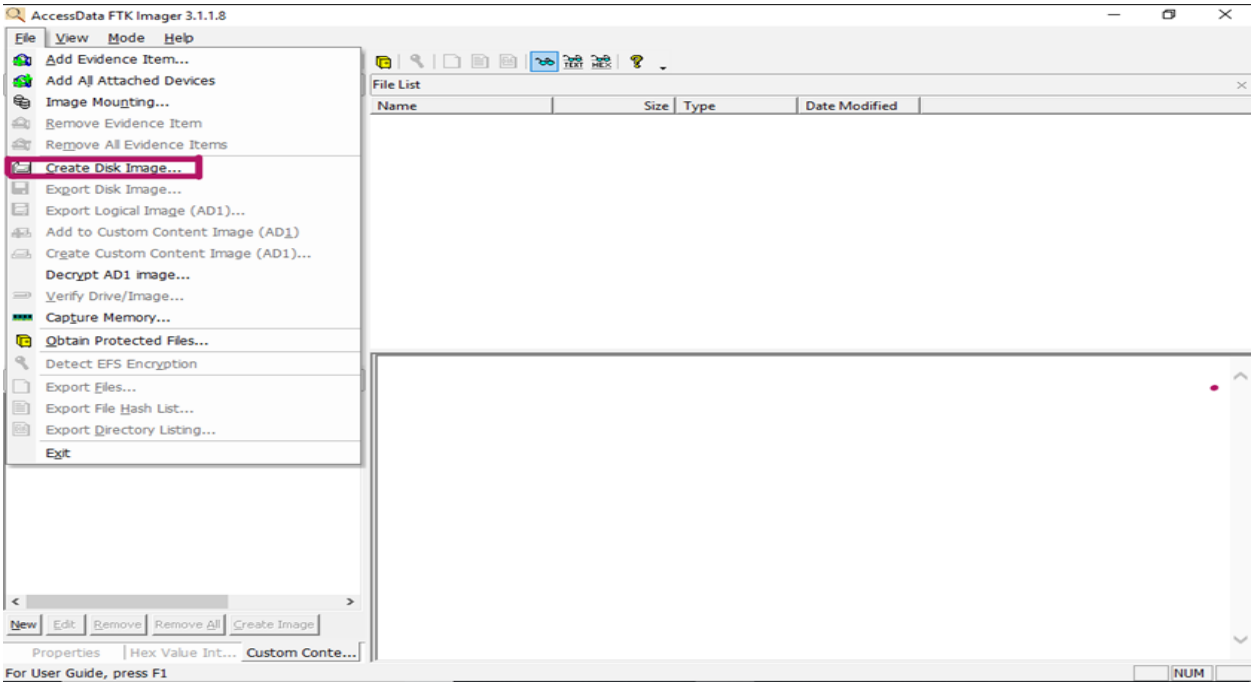
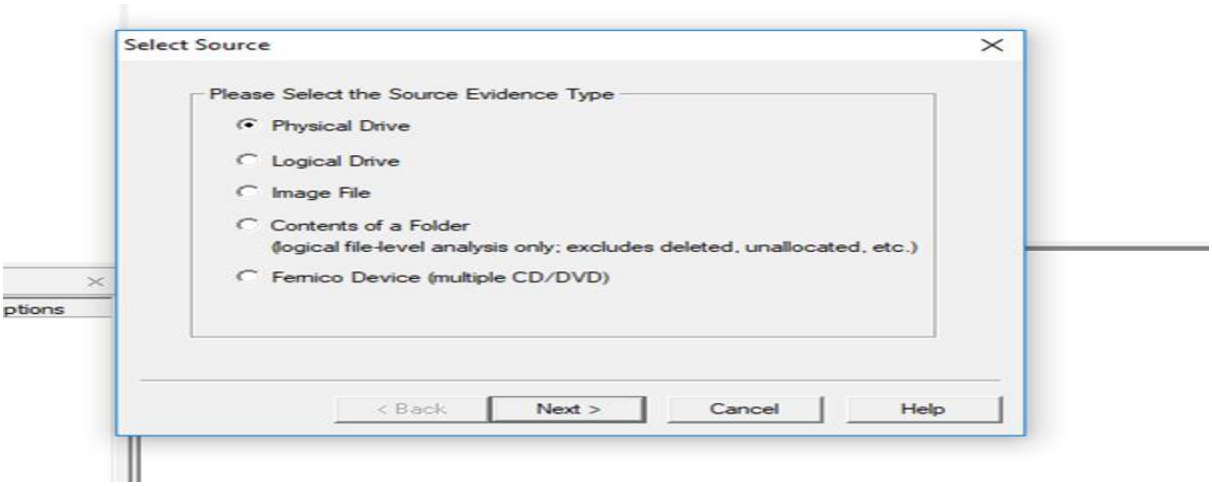
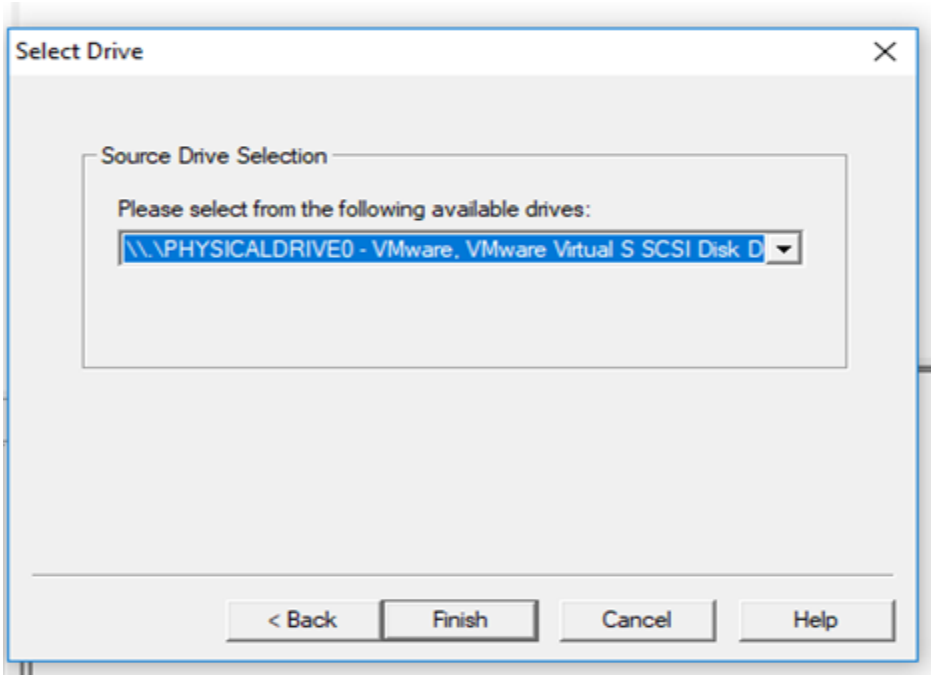


Figure 4 Steps to do HDD Imaging

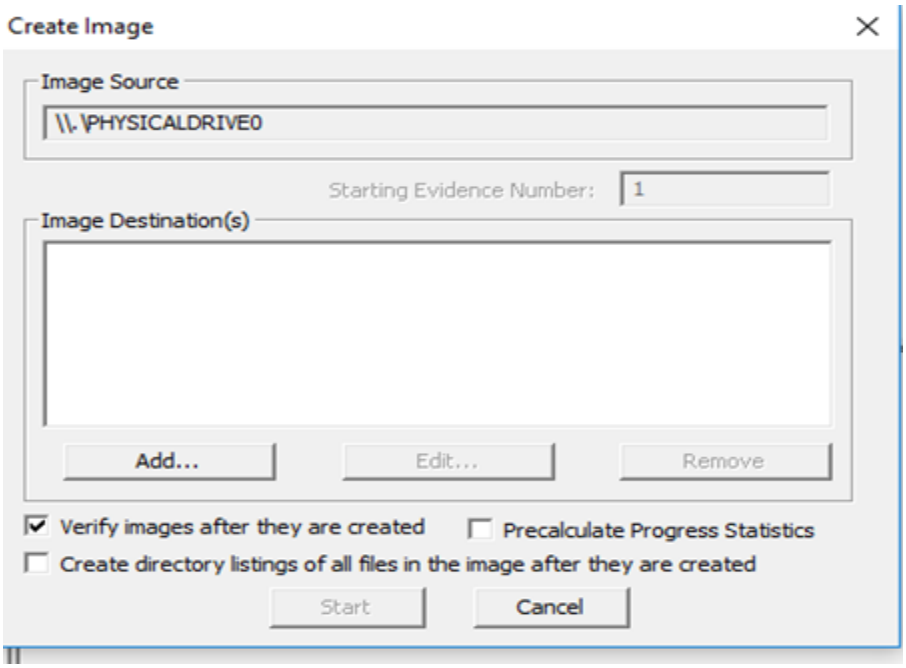
step2:Click Physical Drive



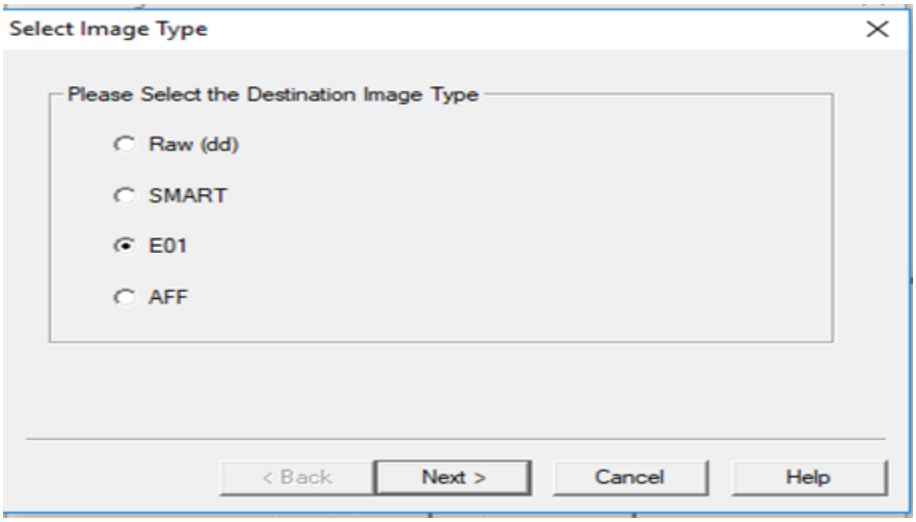
step3:Click Finish in the source drive selection box after selecting Next in the previous window.



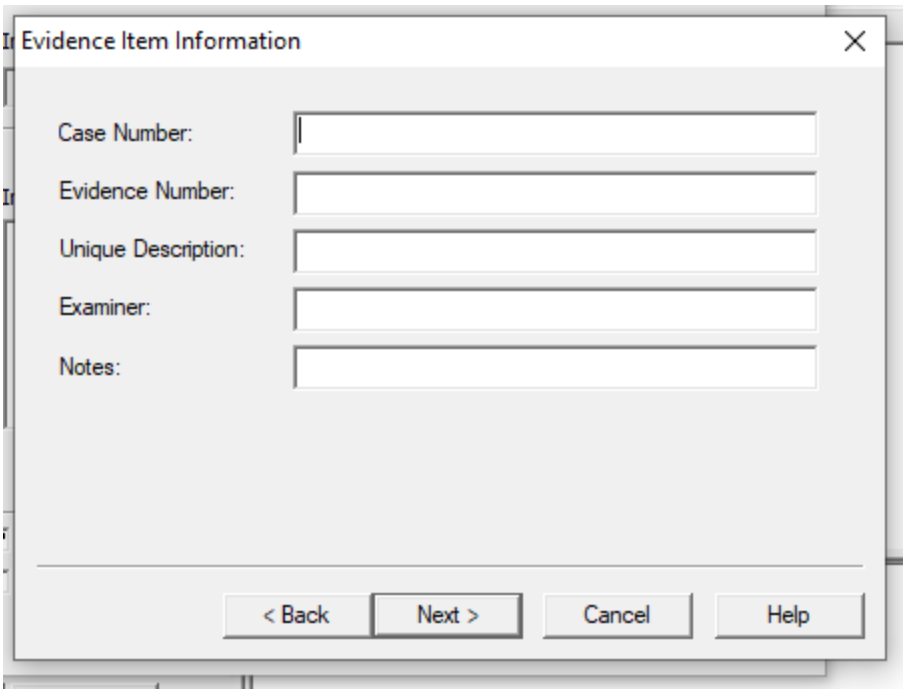
Step 4: An image creation window with an image destination choice will appear once you click "finish" in the previous window. Select the proper location for the data to be stored.



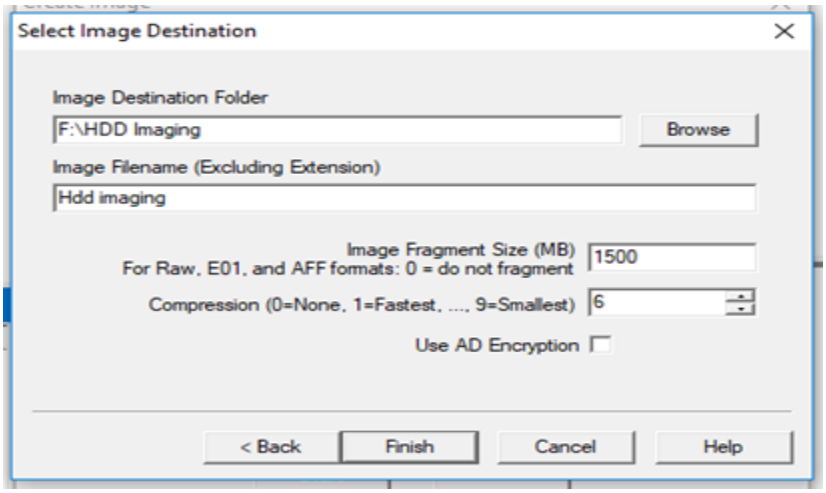
Step 5: After that, a popup titled "Select Image type" will appear. Select E01 to get the evidence in an ordered fashion.



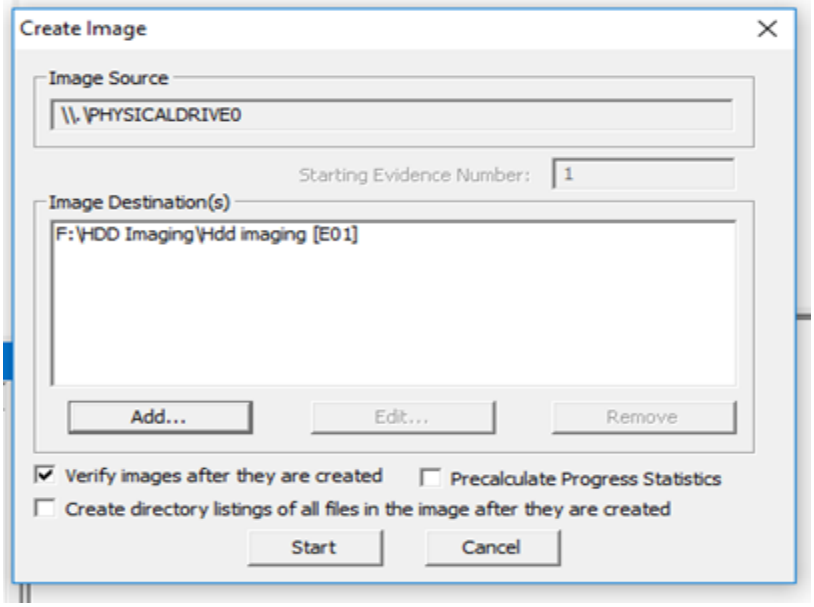
Step 6: Fill up the evidence item information and click next in the window.



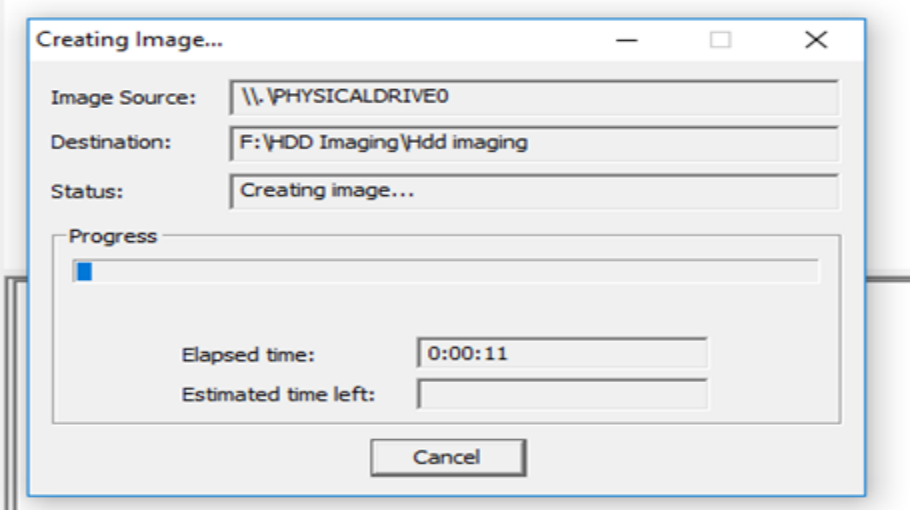
Step 7: The window that appears to choose the picture destination will now reroute. choose a suitable location to save the image file.



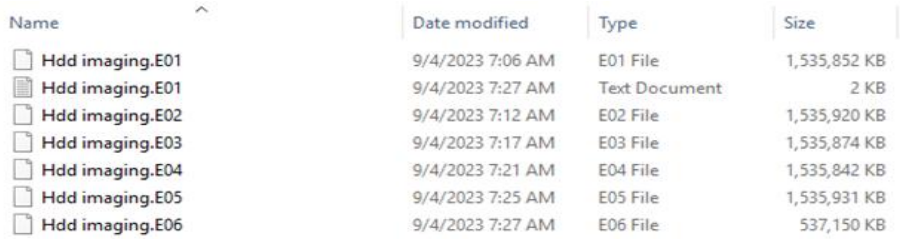
Step 8: After everything is in order, you will see the window shown in Figure 12 and need to click Start.



Step 9: This window will appear when everything is finished, indicating that an image is being created.



Step 10: After the image has been made, it will be saved as the image below illustrates.

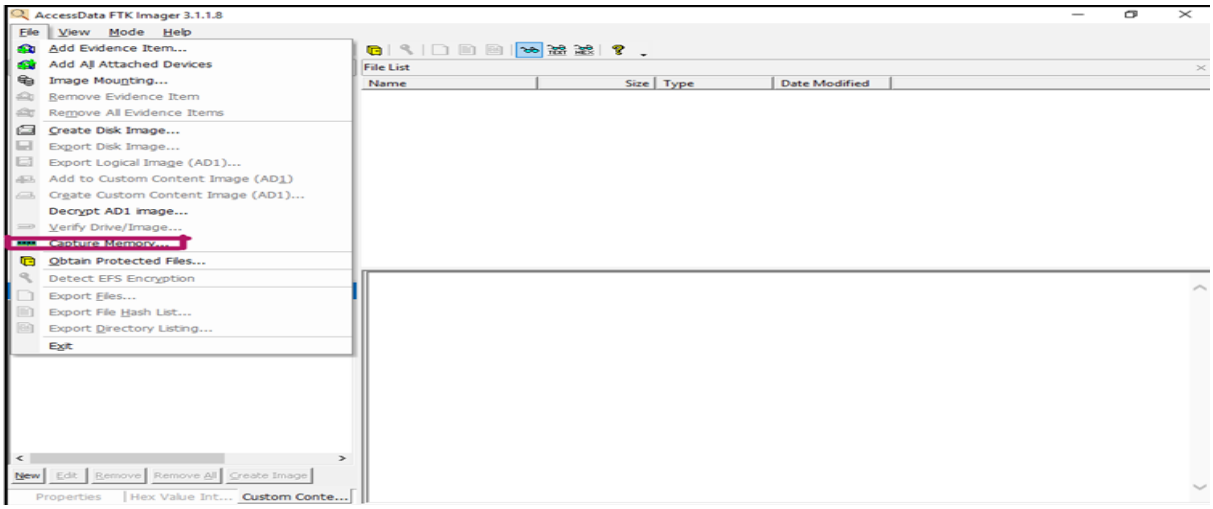


Name	Date modified	Type	Size
Hdd imaging.E01	9/4/2023 7:06 AM	E01 File	1,535,852 KB
Hdd imaging.E01	9/4/2023 7:27 AM	Text Document	2 KB
Hdd imaging.E02	9/4/2023 7:12 AM	E02 File	1,535,920 KB
Hdd imaging.E03	9/4/2023 7:17 AM	E03 File	1,535,874 KB
Hdd imaging.E04	9/4/2023 7:21 AM	E04 File	1,535,842 KB
Hdd imaging.E05	9/4/2023 7:25 AM	E05 File	1,535,931 KB
Hdd imaging.E06	9/4/2023 7:27 AM	E06 File	537,150 KB

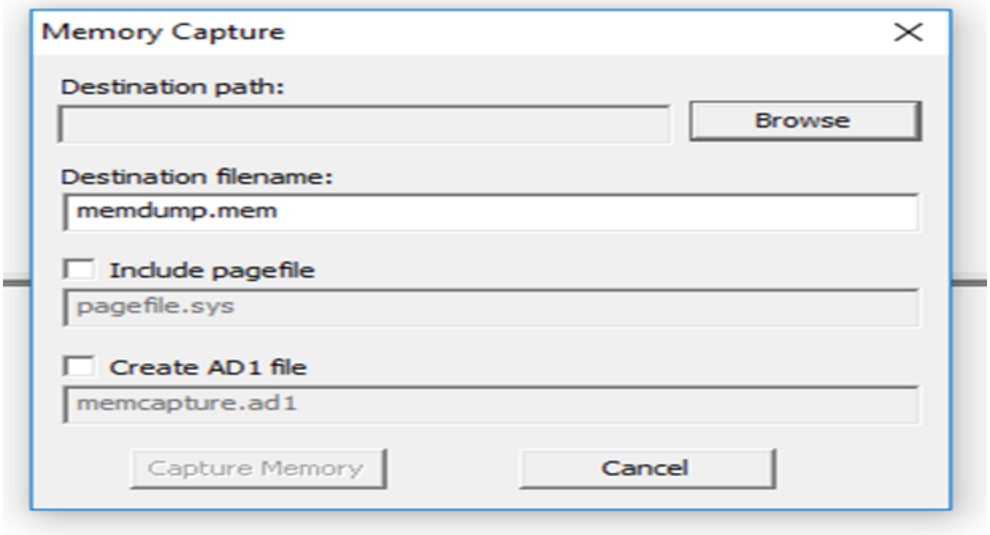
3.2 RAM Capture

RAM data is needed for this infected PC project. Malware detection requires real-time RAM capture of the computer's state. By accessing volatile memory, I can find malware and hidden apps that avoid file-based analysis. RAM grabs reveal network connections, user behavior, including encryption keys and passwords. A comprehensive approach that includes HDD image analysis and RAM capture can reveal malicious activities on the compromised machine.

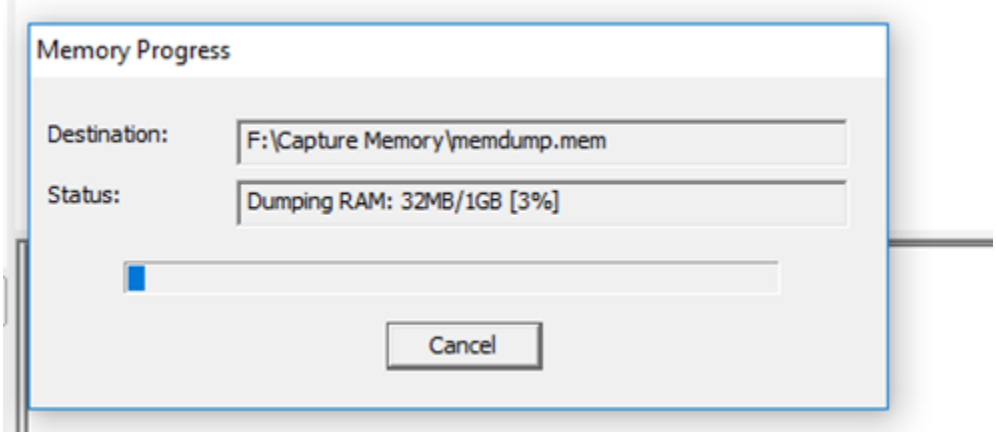
Step 1: open FTK Imager and choose Capture Memory under the File tab.



Step 2: Choose an appropriate destination to store the captured memory.



Step 3: Click "Capture Memory" from the window above after selecting the appropriate place. The window below will then appear, indicating that the picture is being captured.



Step 4: After everything is finished, the window will display "memory captured successfully," indicating that the picture was successfully taken.



4.0 Evidence Analysis

4.1KMSAuto Net.exe

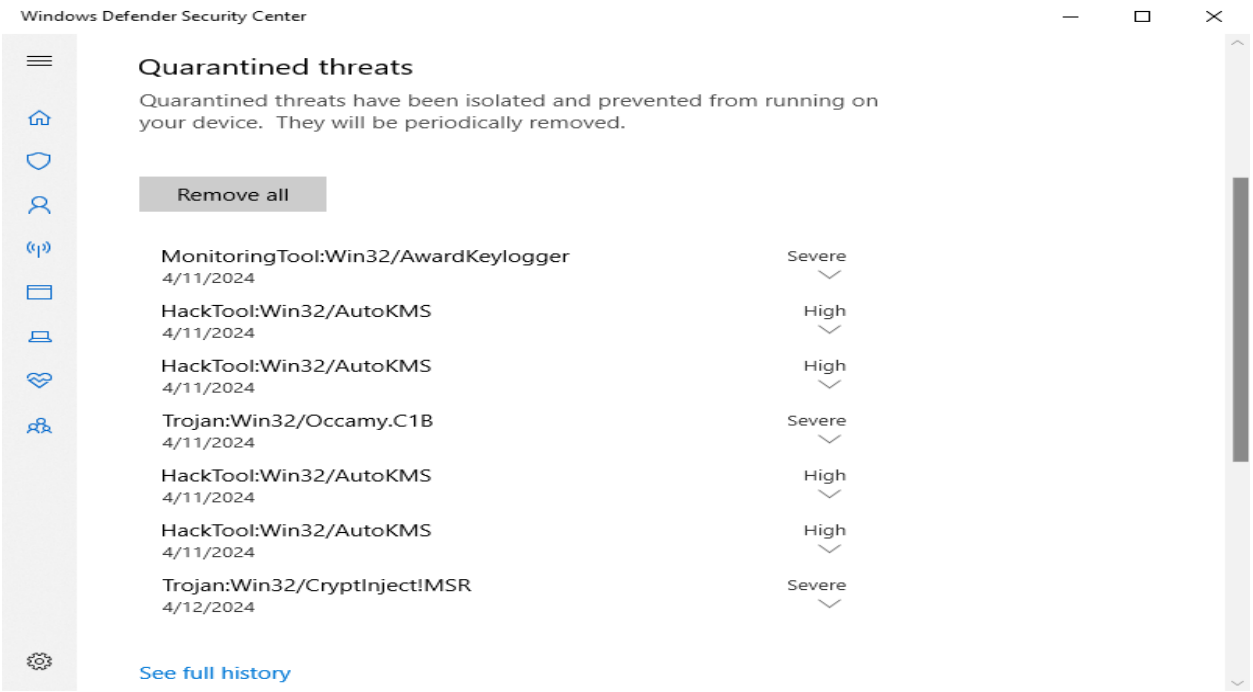


Figure 4.1.1: windows defender found some suspicious files

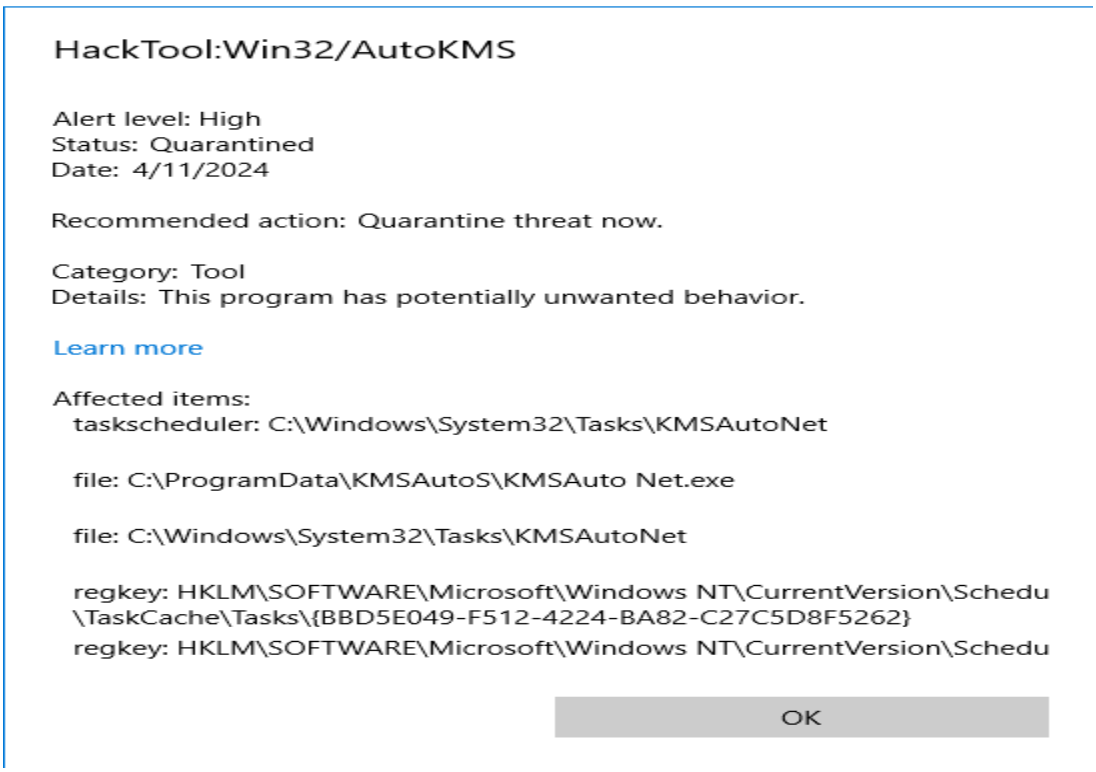


Figure 4.1.2: windows defender found KMSAuto Net.exe has unwanted behavior

Figure 4.1.3: KMSAuto Net.exe detected running in Volatility 3 process list (Windows.pslist)

Following a scan for any suspicious background processes using the Volatility 3 pslist plugin. With a process ID (PID) of 6904, KMSAuto Net.exe is among the dubious processes.

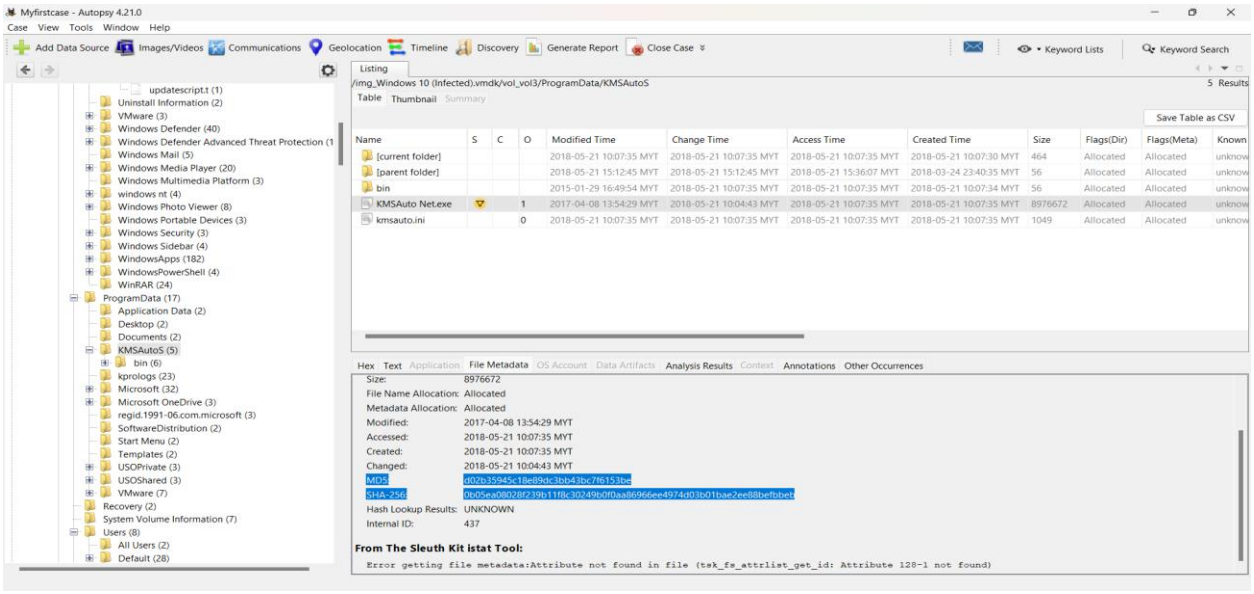


Figure 4.1.4:I found the file Metadata for KMSAuto Net.exe.

The next step is to find KMSAuto Net.exe at "C:\ProgramData\KMSAutoS\KMSAuto Net.exe" in order to file hash from metadata after finishing all of the volatility 3 analysis.

And this the MD5 and SHA-256:

MD5: d02b35945c18e89dc3bb43bc7f6153be

SHA-256: 0b05ea08028f239b11f8c30249b0f0aa86966ee4974d03b01bae2ee88befbbeb

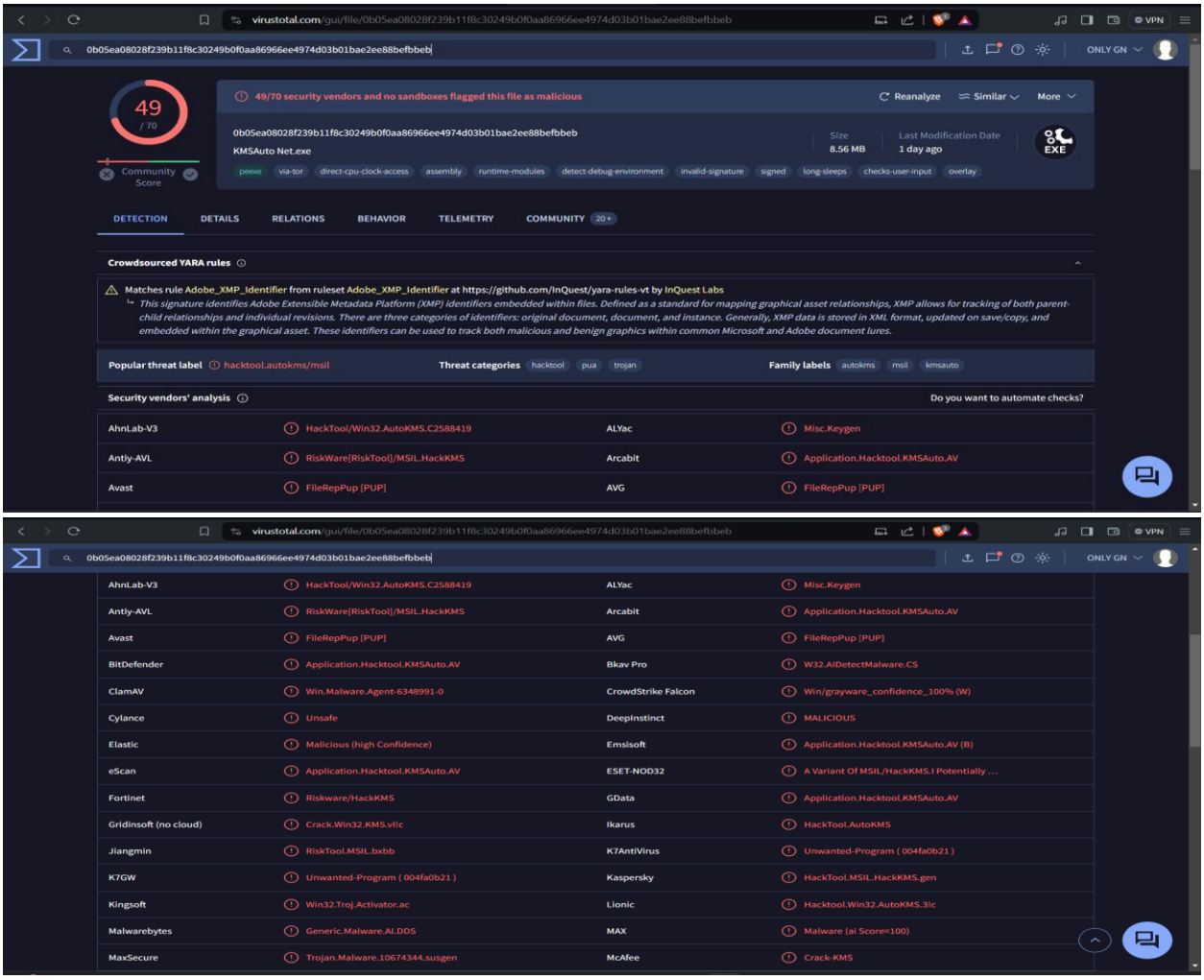


Figure 4.1.5: The Result of the VirusTotal for KMSAuto Net.exe.

We verify that the file is malware by running its unique MD5 "Checksum" against VirusTotal (https://www.virustotal.com/gui/home/search). If the MD5 "Checksum" has already been examined, we can determine the file's genus. Initially, we extracted the MD5 for KMSAuto Net.exe from the file information and sent it to VirusTotal for analysis. The results indicate that KMSAuto Net.exe was identified as malicious.

4.2 DLLInjector v2.exe

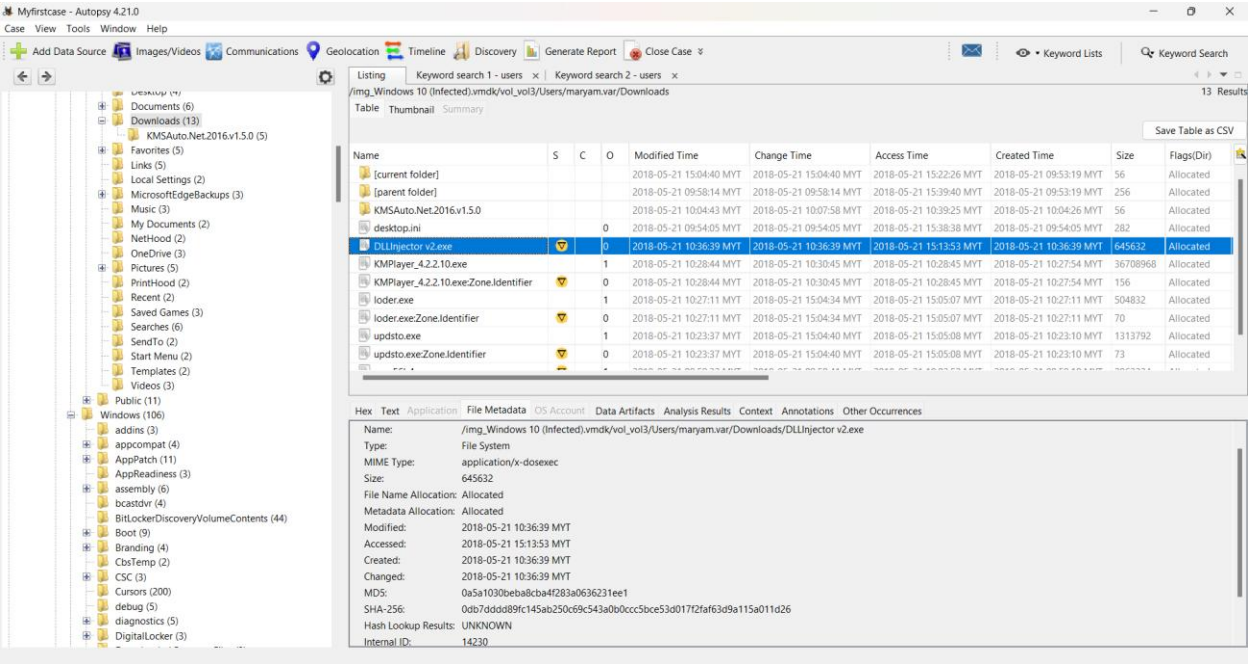


Figure 4.2.1: We found the DLLInjector v2.exe file by using Autopsy



Figure 4.2.2: we found the file Metadata for DLLInjector v2.exe

The next step is to find DLLInjector v2.exe at

"/img_Windows 10 (Infected).vmdk/vol_vol3/Users/maryam.var/Downloads/DLLInjector v2.exe " in order to file hash from metadata after finishing all of the volatility 3 analysis.

This is the MD5 and SHA-256 found:

MD5: 0a5a1030beba8cba4f283a0636231ee1

SHA-256: 0db7dddd89fc145ab250c69c543a0b0ccc5bce53d017f2faf63d9a115a011d26

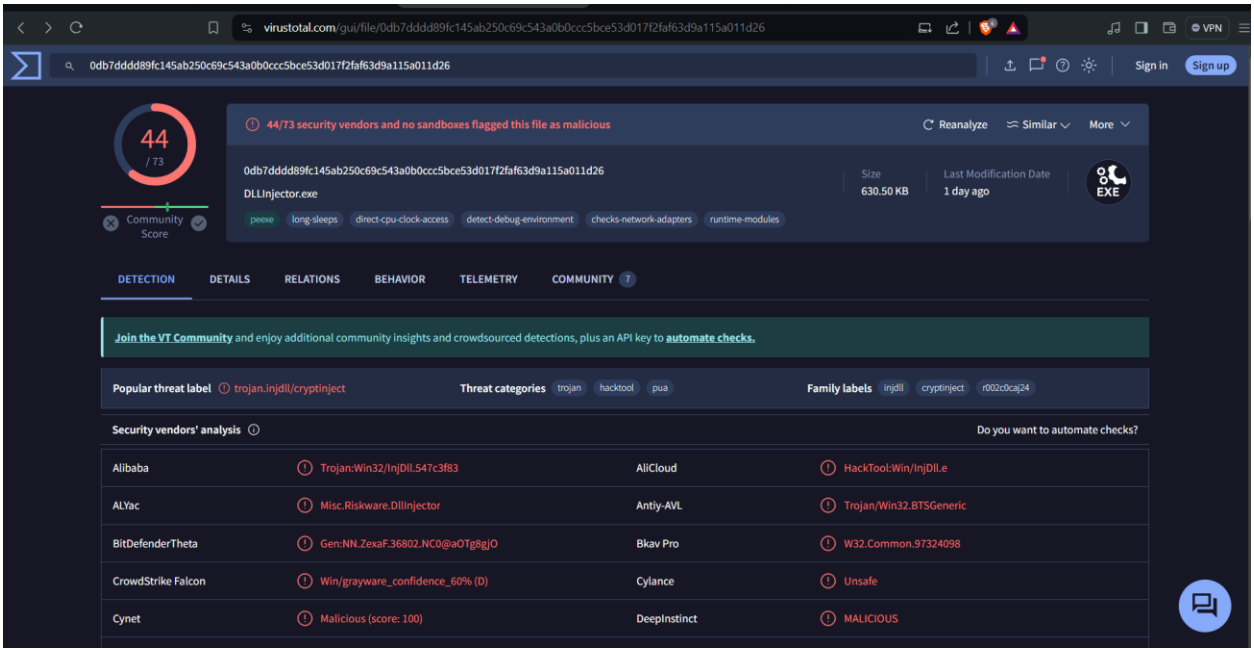


Figure 4.2.3: The Result of the VirusTotal for KMSAuto Net.exe

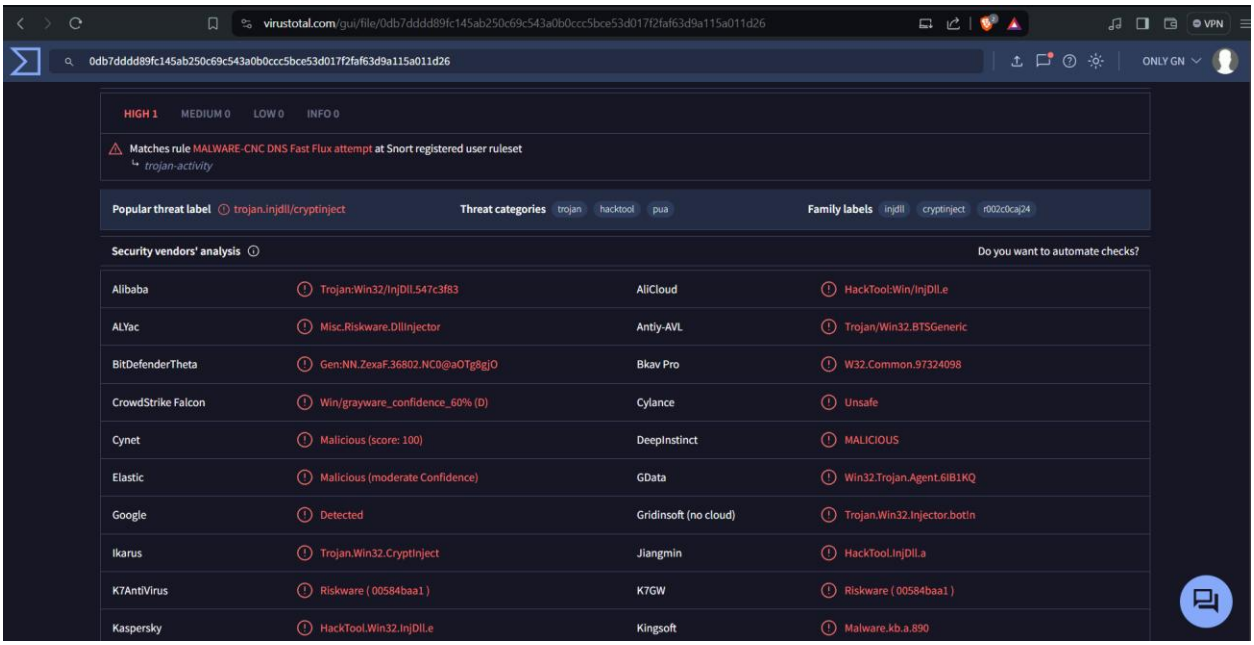


Figure 4.2.4: The Results of the VirusTotal for KMSAuto Net.exe

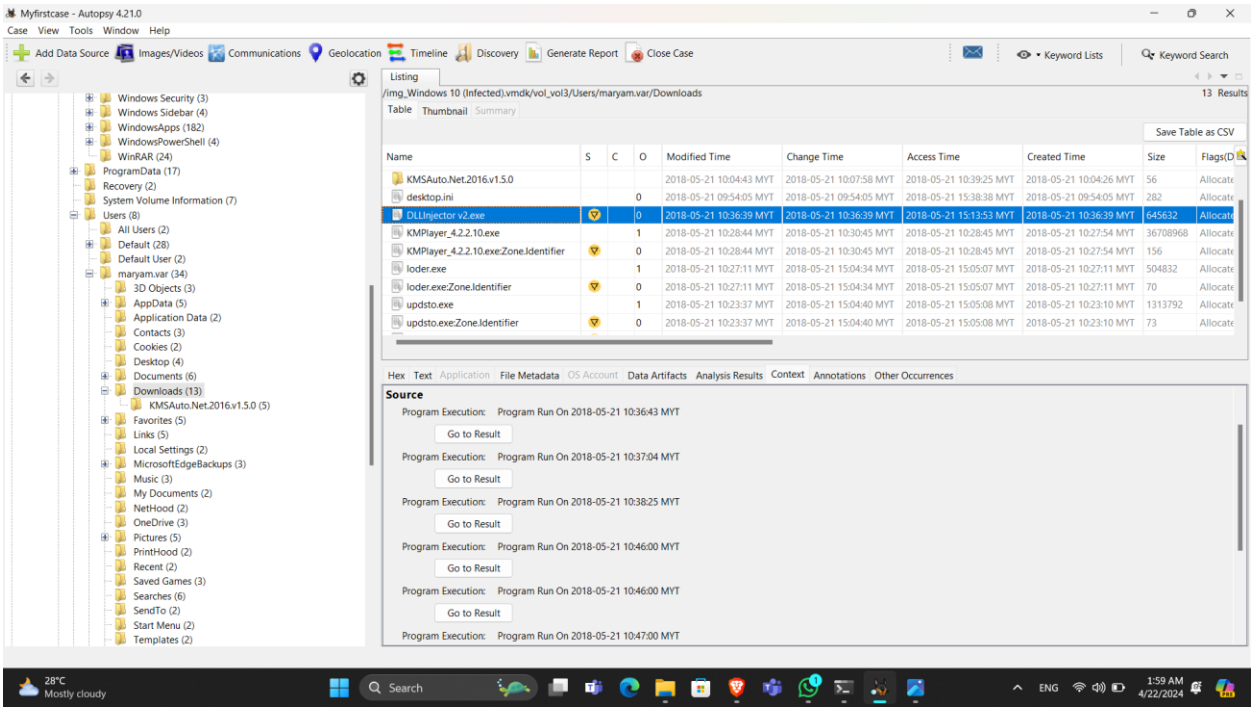


Figure 4.2.4: We found DLLInjector v2.exe file is running in different times

9 executions of the programme are shown when identifying the DLLInjector V2.exe in Autopsy via the context tab

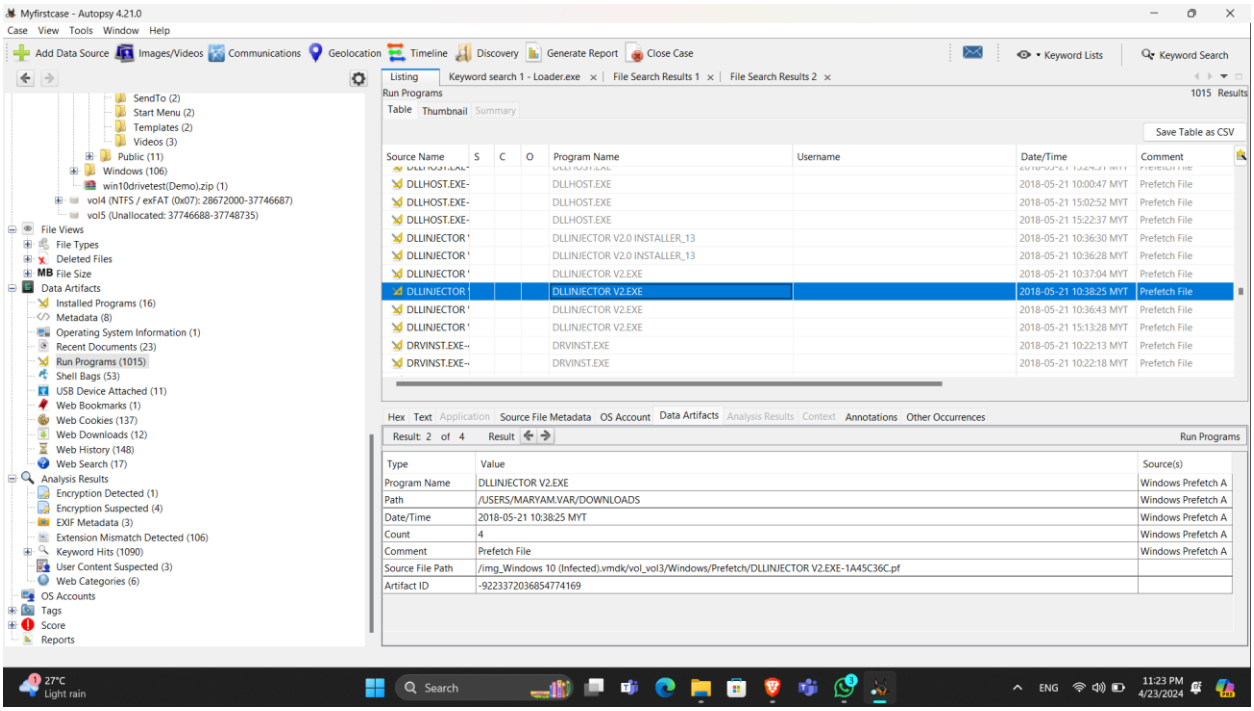


Figure 4.2.5:We found in the run program field that the file is running in the background

4.3 wap.exe

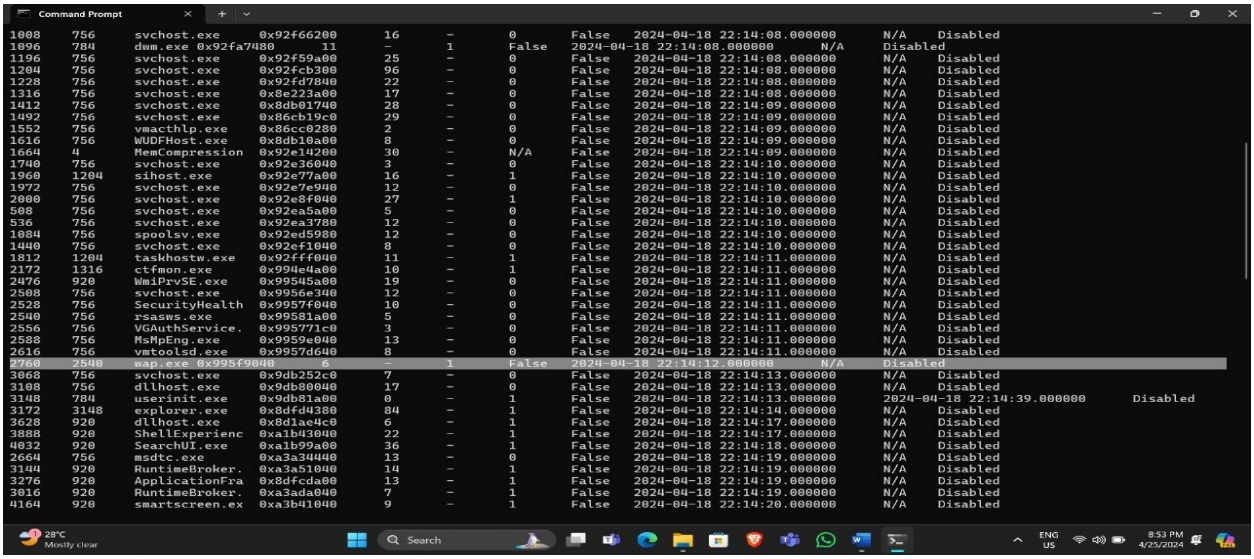


Figure 4.3.1: Volatility 3 process list (Windows.pslist) shows that wap.exe is currently running in the background.

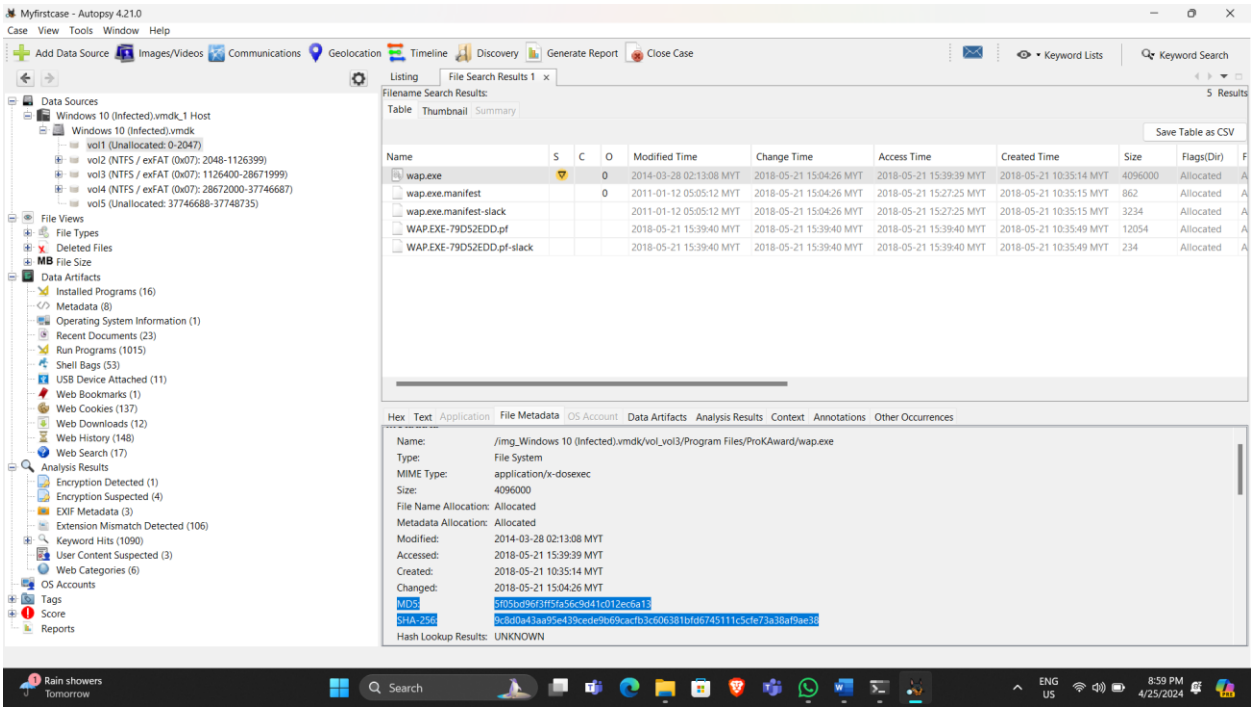


Figure 4.3.2: SHA-256 / MD5 hash value for wap.exe

The next step is to find wap.exe at "/img_Windows 10 (Infected).vmdk/vol_vol3/Program Files/ProKAward/wap.exe" in order to file hash from metadata after finishing all of the volatility 3 analysis.

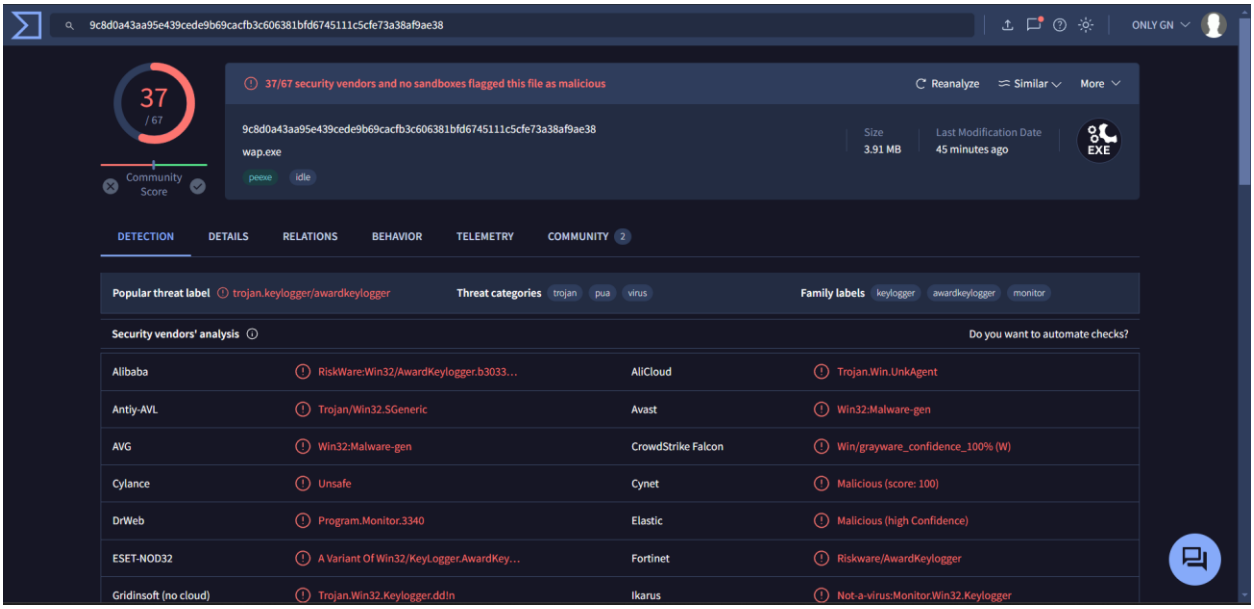


Figure 4.3.3: Results of the suspicious file wap.exe MD5 hash value in VirusTotal

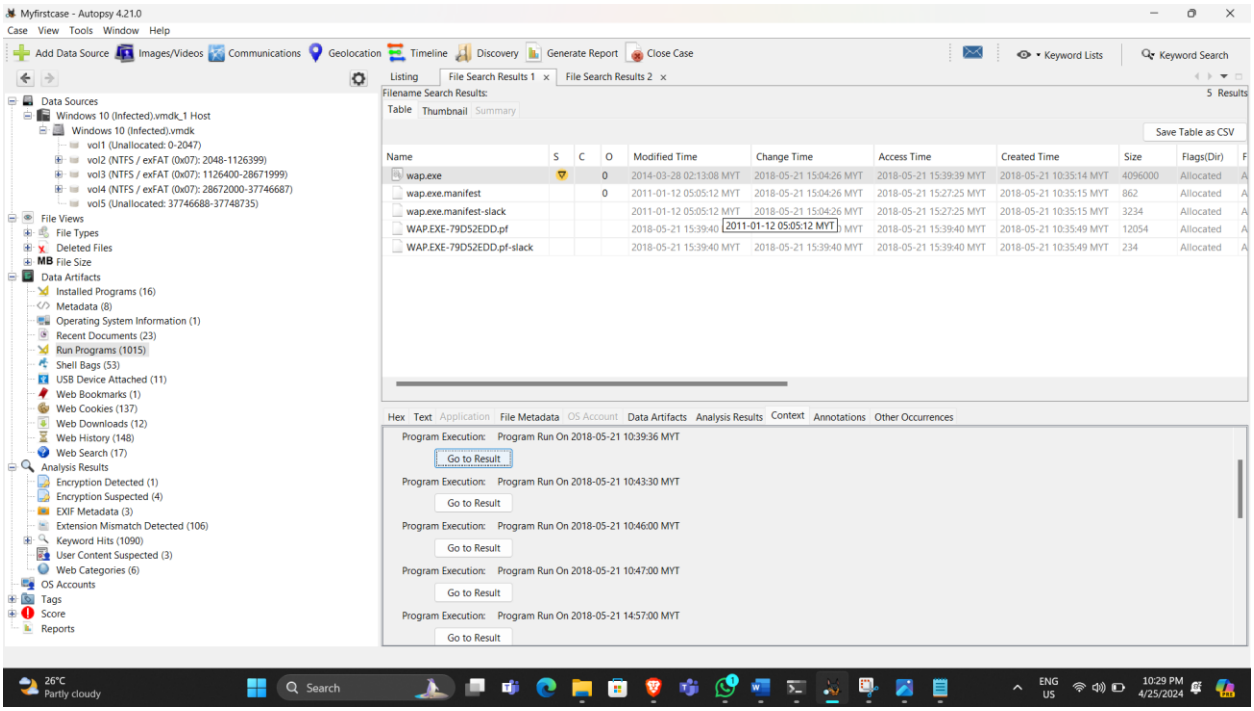


Figure 4.3.4: We found one of the behaviors of Wap.exe file

14 executions of the programme are shown when identifying the wap.exe in Autopsy via the context tab.

5.0 Technical Report

In this study, we use the Volatility 3 framework and autopsy to conduct a forensic investigation and analyze three questionable processes. DLLInjector v2.exe, wap.exe, and KMSAuto Net.exe are the processes that have been examined. We thoroughly investigated every process, gathering metadata including SHA-256 and MD5 hashes among other things. We then used VirusTotal to confirm that they were malicious and recorded their actions.

5.1 Analysis of KMSAuto Net.exe

Process Details:

Name: KMSAuto Net.exe

Path: C:\ProgramData\KMSAutoS\KMSAuto Net.exe

PID: 6904

File Hashes:

MD5: d02b35945c18e89dc3bb43bc7f6153be

SHA-256: 0b05ea08028f239b11f8c30249b0f0aa86966ee4974d03b01bae2ee88befbbeb

VirusTotal Analysis:

MD5 Checksum: d02b35945c18e89dc3bb43bc7f6153be

Result: Identified as malicious

Behaviors:

1. Provides additional service for the system.
2. Modifies registry keys.
3. Adjusts the Windows firewall.
4. Terminates active services.

5.2 Analysis of DLLInjector v2.exe

Process Details:

Name: DLLInjector v2.exe

Path: /img_Windows 10 (Infected).vmdk/vol_vol3/Users/maryam.var/Downloads/DLLInjector v2.exe

File Hashes:

MD5: 0a5a1030beba8cba4f283a0636231ee1

SHA-256: 0db7dddd89fc145ab250c69c543a0b0ccc5bce53d017f2faf63d9a115a011d26

Behaviors:

1. Uses encryption for injection.
2. Engages in data theft.
3. Facilitates remote entry with tenacity.
4. 9 executions of the programme are shown when identifying the DLLInjector V2.exe in Autopsy via the context tab

5.3 Analysis of wap.exe

Process Details:

Name: wap.exe

Path: /img_Windows 10 (Infected).vmdk/vol_vol3/Program Files/ProKAward/wap.exe

File Hashes:

MD5: 5f05bd96f3ff5fa56c9d41c012ec6a13

SHA-256: 9c8d0a43aa95e439cede9b69cacfb3c606381bfd6745111c5cfe73a38af9ae38

Behaviors:

1. Captures keystrokes.
2. Accesses clipboard contents.
3. Takes screenshots.
4. Records application usage.
5. 14 executions of the programme are shown via the context tab

Conclusion

KMSAuto Net.exe, DLLInjector v2.exe, and wap.exe show malicious characteristics suggestive of possibly hazardous activity, according to the analysis of the suspicious processes. It is essential to take the necessary steps to reduce the risks that these procedures offer, such as eliminating them from the system and putting security measures in place to stop occurrences like this from happening again.

6.0 Conclusion

Finally, this digital forensics research on compromised PCs and virus detection was intriguing. This study evaluates and offers expert findings. I studied digital forensics fundamentals.

My experience began with tactical data collection using forensic tools. Methodically employing each instrument's capabilities increased RAM and HDD data analysis. For its flexibility, simplicity of use, picture-making, and hash verification, FTK Imager was trusted. Autopsy and Volatility3, trusted data analysts, delivered digital evidence narratives.

My study used FTK Imager's versatility to demonstrate tool selection's relevance in dynamic situations. Real-time RAM communication evidence finishes. Integrating user context, covert network activity, and memory evidence reveals the compromised system.

Autopsy and Volatility revealed a flawed system3. Using its MD5 hash, we could peek inside the system during autopsy. Operating system and installation date were revealed. Hidden user

activity, wiped data, and security issues were detected in files. System digital interactions depend on networking.

Volatility3 used memory forensics to find odd activity, parent-child connections, and network linkages. These findings strongly suggested malware, remote command, and illicit activity.

My research, including our analysis, shows the compromised PC. Malicious process spoofing, unlawful process injections, anomalous process affiliations, . Abnormalities threaten data security, system integrity, and digital defense.

I want to increase cybersecurity and solve digital puzzles. We plan and focus. Every digital forensics problem solved puts us closer to a safer future, therefore the hunt continues.

7.0 References

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